

10 Gentle Knee Exercises for Arthritis — Physio-Approved

Ten safe, effective exercises for arthritic knees, designed by physiotherapists. Strengthen your knees, reduce pain, and improve mobility — all at home.

By **Sarah Jennings, HCPC Registered Physiotherapist** · Reviewed by **Sarah Jennings, HCPC Registered Physiotherapist** (PH123456, CSP Member) · 2026-06-16 · livingwitharthritis.org.uk

Clinically reviewed by **Sarah Jennings**, HCPC Registered Physiotherapist (PH123456) and CSP member. Last updated: June 2026.

"If I move it, it will hurt more." We hear this a lot. It is the most common worry for people with arthritic knees — and it is also the biggest myth. Gentle, regular movement is the best thing you can do for painful knees. Strengthening the muscles around the joint provides natural support, reduces strain, and — over time — lessens pain.

The exercises below are the same ones NHS physiotherapists recommend. They are gentle, equipment-free, and you can do them at home.

Why Exercise Helps Arthritic Knees (NICE Guidance)

The National Institute for Health and Care Excellence (NICE) is clear: exercise should be the **first treatment** offered for knee osteoarthritis. Not medication. Not surgery. Exercise. Here is what the evidence shows: strengthening the quadriceps and glutes reduces pressure on the knee joint; regular movement nourishes cartilage and keeps joints lubricated; exercise improves balance and stability, reducing fall risk; and it helps with weight management — every kilogram lost reduces force through your knee by roughly 4 kilograms.

Before You Start — Safety Tips



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- Gentle is the goal. You should feel a stretch or mild muscle ache, not sharp pain. Sharp pain means stop.
 - Start small. Do 5 repetitions of each exercise to begin, building to 10–15 as you feel stronger.
 - Warm up. A warm bath or heated wheat bag on your knee for 5 minutes before exercising can ease stiffness.
 - Breathe. Don't hold your breath. Breathe out as you lift or push.
 - Speak to a health professional if you have had recent surgery, a fall, or other medical conditions.
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10 Knee-Strengthening Exercises

1. Seated Knee Straightening (Quad Sets)

Best for: Activating the thigh muscle without stressing the joint. Sit on a chair with your feet flat on the floor. Slowly straighten one leg until it is parallel to the floor. Hold for 5 seconds. Slowly lower it back down. Repeat on the other leg.

2. Sit-to-Stand

Best for: Functional strength — getting out of chairs. Sit on a firm chair with your feet hip-width apart. Lean forward slightly and stand up, using your legs (not your hands on your thighs). Slowly lower yourself back down. If this is too hard, start from a higher chair or use armrests.

3. Static Quadriceps Contraction

Best for: Waking up the quadriceps on days when movement feels too painful. Sit or lie with your leg straight out. Tighten the muscle at the front of your thigh by pressing the back of your knee down towards the floor. Hold for 5–10 seconds, then relax. No visible movement needed — it is all about the squeeze.

4. Straight Leg Raise

Best for: Strengthening the quadriceps and hip flexors. Lie on your back with one leg bent (foot flat) and the other leg straight. Tighten the thigh muscle of the straight leg and lift it to the height of the bent knee. Hold for 3–5 seconds, then lower slowly. Keep your lower back pressed gently into the floor.

5. Glute Bridge

Best for: Strengthening the glutes, which support the knees. Lie on your back with knees bent and feet flat, hip-width apart. Squeeze your buttocks and lift your hips until your body forms a straight line from knees to shoulders. Hold for 5 seconds, then lower slowly.

6. Seated Hamstring Curl

Best for: Strengthening the back of the thigh. Sit on a chair with a smooth floor under your feet. Keep one foot planted. Slowly slide the other heel back under the chair, bending your knee. Hold for 3 seconds, then slide it forward again. For added resistance, place a towel under your heel.

7. Calf Raises

Best for: Improving ankle and knee stability. Stand holding the back of a sturdy chair. Rise up onto your toes, lifting your heels as high as is comfortable. Hold for 3 seconds, then lower slowly.

8. Clamshell

Best for: Strengthening the hip abductors — vital for knee alignment. Lie on your side with knees bent at roughly 45 degrees and hips stacked. Keeping your feet together, lift your top knee as high as you can without rolling your pelvis backwards. Hold for 3 seconds, then lower. You should feel it in the side of your hip.

9. Wall Squat (Mini Squat)

Best for: Building leg strength without deep bending. Stand with your back against a wall, feet hip-width apart and about 30cm from the wall. Slowly slide down, bending your knees no more than 45 degrees. Hold for 5 seconds, then slide back up.

10. Heel Slides

Best for: Gently improving knee bend range of motion. Sit or lie with your legs out straight. Slowly slide one heel towards your buttock, bending your knee as far as is comfortable. Hold for 3 seconds, then slide it back.

How Often Should You Exercise?



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Aim for: **daily** — exercises 1 (quad sets) and 3 (static contractions); **3–4 times per week** — the full set of all 10 exercises; **2–3 times per week** — a short walk, swim, or cycle. Consistency matters more than intensity. Ten minutes every other day beats one heroic hour once a week.

What If Exercise Hurts?

Good sensations: a dull ache or muscle tiredness, stiffness that eases as you warm up, mild soreness that fades within 24 hours. **Warning signs:** sharp stabbing pain during an exercise, pain that gets worse with each repetition, swelling lasting more than 2 hours after exercising, any pain that persists or worsens over several days. **Sharp pain is your body saying no.**

Q&A

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General information only — not medical advice. Consult your GP or specialist before changing medication or starting a new programme. Emergency: 999. NHS advice: 111. Living With Arthritis · livingwitharthritis.org.uk